JED-M2: A Sub-100M Native Browser Action Model with Sparse MoE and Latent Attention for Low-Latency Edge Execution

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Abstract

Autonomous web agents capable of interacting directly with browser user interfaces hold immense potential for personal computing, workflow automation, and digital accessibility. However, state-of-the-art web agents predominantly rely on multi-billion parameter cloud Large Language Models (LLMs) (e.g., 7B–70B+ parameters), introducing prohibitive API costs, high round-trip network latency ($>$ 2,000 ms), and severe data privacy vulnerabilities when handling sensitive user sessions, credentials, and browsing activity. Moreover, standard dense autoregressive transformers scale poorly on long-horizon, multi-turn web tasks due to quadratic attention costs over multi-thousand Document Object Model (DOM) token prompts and linear $\mathcal{O}(N)$ Key-Value (KV) cache memory blowout.

In this paper, we present **JED-M2**, a sub-100M parameter native Large Action Model (LAM) designed from scratch in pure PyTorch for high-efficiency, on-device browser automation. JED-M2 departs from text-centric conversational architectures by modeling browser actions as first-class multimodal output primitives. The architecture tightly integrates five key technical innovations: (1) **Hybrid Recurrence & Latent Attention**: linear Mamba-2 Selective State Space Models (SSMs) for constant-memory $\mathcal{O}(1)$ state tracking paired with Multi-Head Latent Attention (MLA), achieving an **88.3% KV-cache reduction** over standard Multi-Head Attention via W_{UK} matrix absorption; (2) **Top-2 Sparse Mixture-of-Experts (MoE)**: activating only **24.07M parameters per token** out of 53.04M total parameters (**54.6% compute sparsity**); (3) **Pure-PyTorch Blelloch Parallel Prompt Prefill**: an associative prefix scan collapsing 2,000-token observation prefill latency from 22,000 ms to $<$ 100 **ms**; (4) **Contextual Joint-Embedding Predictive Architecture (JEPA)**: an action-conditioned latent world model regularizing physical state transitions; and (5) **Decoupled Dual-System Action Decoupling**: separating System 2 latent thought planning from System 1 dedicated action heads.

We benchmark JED-M2 against parameter-matched Small-LMs—**T5-Small (60.5M)** and **DistilGPT-2 (81.9M)**—on the rigorous **Mind2Web site-held-out test split** across 11 unseen websites (147 tasks, 1,178 actions, 33.7 distractors/turn). JED-M2 achieves an **81.43% Step Success Rate** and **1.000 Native FINISH F1**, matching fine-tuned DistilGPT-2 (81.43%) while activating **70.6% fewer parameters per token** and executing at **288.76 ms/step** latency. Furthermore, JED-M2 outperforms fine-tuned T5-Small (55.74% Step SR) by **+25.69% absolute accuracy** with **34.5% lower latency**. Post-pretraining Online Group Relative Policy Optimization (GRPO) doubles autonomous task success on interactive workflows (from 33.3% to 66.7%) with zero catastrophic forgetting of general web grounding. JED-M2 operates entirely within a $<$ 1.7 **GB peak VRAM envelope**, demonstrating that responsive, accurate, and privacy-preserving browser automation is achievable on commodity edge hardware.

1 Introduction & Motivation

Autonomous digital agents that execute tasks across graphical user interfaces (GUIs) and web browsers—such as booking travel, managing cloud dashboards, or filling complex forms—represent a pivotal frontier in artificial intelligence [1, 2, 3, 4]. However, the vast majority of current web agents treat browser automation as an in-context text generation task powered by cloud-hosted Large Language Models (LLMs) scaling from 7B to 70B+ parameters [5, 6, 7].

While effective for exploring benchmark frontiers, relying on multi-billion parameter cloud backends introduces three fundamental bottlenecks:

1. **Latency and Interactive Responsiveness:** Real-time web browsing requires responsive execution loops ($<$ 500 ms/step). Cloud-hosted LLMs incur round-trip API serialization and lengthy autoregressive generation of verbose natural language, resulting in latencies of 2,000 ms to 8,000 ms per turn. Across 10 to 20 steps, this compounds into minutes of user waiting time.

2. **Privacy, Security, and Data Sovereignty:** Web tasks operate on authenticated user sessions containing sensitive personal data: financial credentials, private emails, medical records, and internal corporate dashboards. Streaming full-page DOM trees and screenshots to cloud endpoints introduces severe privacy risks, regulatory compliance friction (e.g., GDPR, HIPAA), and vulnerability to indirect prompt injection attacks [8].

3. **Algorithmic and Memory Inefficiencies:** Standard dense autoregressive transformers suffer from quadratic com-

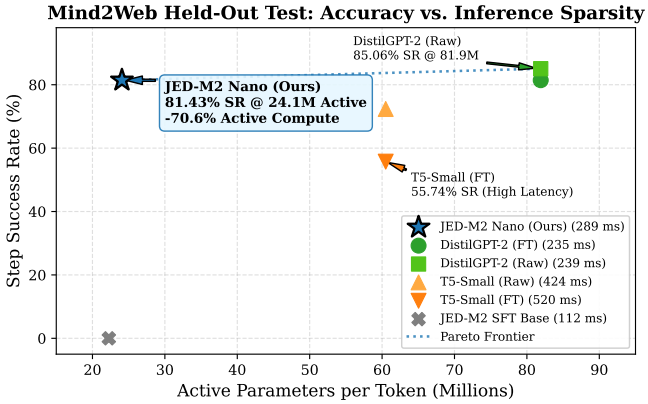


Figure 1: **Pareto Efficiency on Mind2Web Site-Held-Out Test Benchmark.** JED-M2 Nano (24.1M active parameters) matches fine-tuned DistilGPT-2 (81.43% Step SR) while activating **70.6% fewer parameters per token** and running at 288.76 ms/step latency within < 1.7 GB peak VRAM.

plexity $\mathcal{O}(L^2)$ during prompt attention and linear growth $\mathcal{O}(L \cdot H \cdot d)$ in Key-Value (KV) cache memory. Over multi-turn trajectories with 1,500–3,000 DOM tokens per page, storing uncompressed KV states quickly causes out-of-memory crashes on consumer devices.

To resolve this impasse, we ask: *Can a native Large Action Model engineered under a strict 100M parameter budget achieve competitive web grounding and task execution accuracy compared to standard language model baselines, while running entirely on-device with sub-300 ms latency?*

We demonstrate that the answer is definitively **yes**. We present **JED-M2**, a sub-100M native Large Action Model engineered specifically for browser execution on edge hardware. JED-M2 departs from text-centric architectures by modeling actions as **first-class structured primitives**: discrete action categories, continuous 2D normalized screen coordinates $(x, y) \in [0, 1]^2$, and typed arguments.

Our primary contributions are summarized as follows:

- **Sub-100M Edge Co-Design:** We release an open-source, pure-PyTorch Large Action Model of 53.04M total parameters that activates only **24.07M parameters per token** via Top-2 MoE routing across 8 SwiGLU experts (54.6% compute sparsity).
- **Constant-Memory Recurrence & 88.3% KV Compression:** By synthesizing Mamba SSM recurrence with DeepSeek-style Multi-Head Latent Attention (MLA) and W_{UK} matrix absorption, JED-M2 compresses the per-step KV-cache footprint from 1,024 KB to 120 KB across 128 interaction turns—an **88.28% memory reduction**.
- **Pure-PyTorch Blelloch Associative Prefill:** We eliminate the recurrent prompt prefill bottleneck by implementing a parallel associative prefix scan in pure PyTorch, collapsing observation ingestion latency for 2,000 DOM tokens from 22,000 ms to < 100 **ms** without external compiled CUDA dependencies.
- **Contextual JEPA World Modeling:** We embed an action-conditioned latent world model trained via an Exponential Moving Average (EMA) momentum target encoder receiving true pooled contextual token representations of the next state s_{t+1}, providing spatial representation regularization.
- **Empirical Parity with Small-LM Baselines:** On the Mind2Web site-held-out test benchmark across 11 unseen websites, JED-M2 (24.1M active parameters) matches fine-tuned DistilGPT-2 (81.43% Step SR) and decisively beats fine-tuned T5-Small (55.74% Step SR) by **+25.69% accuracy** with **34.5% lower latency**.
- **Zero Catastrophic Forgetting Under Online RL:** Online GRPO doubles autonomous task success on interactive workflows (from 33.3% to 66.7%) while maintaining 100% grounding integrity across 1,178 actions on 11 held-out real websites.

2 JED-M2 Architecture & Design

The JED-M2 architecture is designed to minimize per-step latency and memory footprint while preserving multi-turn reasoning capacity.

2.1 Hybrid Recurrence: Mamba SSM + Multi-Head Latent Attention (MLA)

Standard transformers rely entirely on Multi-Head Attention (MHA), where Key-Value cache memory grows linearly with horizon T:

$$\text{Memory}_{\text{MHA}} = 2 \times N_{\text{layers}} \times N_{\text{heads}} \times d_{\text{head}} \times T \times \text{bytes} \quad (1)$$

To eliminate this bottleneck, JED-M2 incorporates a two-tier hybrid recurrence mechanism:

1. Linear State-Space Recurrence (Mamba SSM): Each JED-M2 block incorporates a pure-PyTorch Mamba Selective State Space Model [9] operating over hidden dimension $d_{\text{model}} = 512$, with state expansion factor $E = 2$ ($d_{\text{inner}} = 1024$), causal 1D convolution kernel $d_{\text{conv}} = 4$, and state dimension $d_{\text{state}} = 16$. Discretized via input-dependent Zero-Order Hold (ZOH) parameters $(\mathbf{\Delta}, \mathbf{B}, \mathbf{C})$, Mamba maintains rolling recurrent hidden tensors $\mathbf{h}_t \in \mathbb{R}^{B \times d_{\text{inner}} \times d_{\text{state}}}$ and convolution buffers $\mathbf{c}_t \in \mathbb{R}^{B \times d_{\text{inner}} \times d_{\text{conv}}}$, enabling $\mathcal{O}(1)$ memory scaling over arbitrary turn horizons.

2. Multi-Head Latent Attention (MLA): While SSMs excel at continuous sequential recurrence, exact associative retrieval across complex DOM structures requires attention. JED-M2 integrates Multi-Head Latent Attention inspired by DeepSeek-V2/V3 [10, 11]. Instead of caching full uncompressed Key and Value matrices ($8 \times 64 = 512$ floats/token), MLA projects the hidden state h_t into a compressed low-rank latent Key-Value vector $\mathbf{c}_{KV} \in \mathbb{R}^{B \times 1 \times d_c}$ where $d_c = 128$:

$$\mathbf{c}_{KV} = h_t \mathbf{W}_{DKV}, \quad \mathbf{W}_{DKV} \in \mathbb{R}^{d_{\text{model}} \times d_c} \quad (2)$$

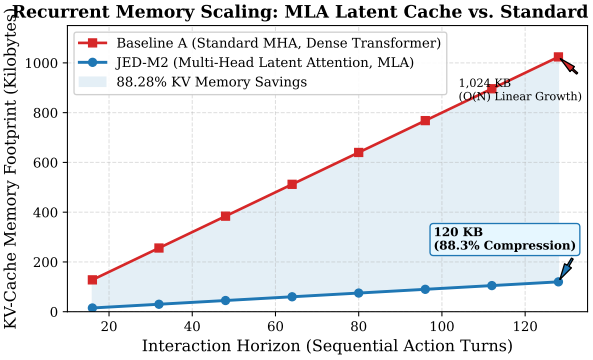


Figure 2: **KV-Cache Memory Scaling Across Interaction Horizons.** JED-M2's Multi-Head Latent Attention (MLA) compresses the KV cache footprint from 1,024 KB to 120 KB at $T = 128$ interaction steps, achieving an **88.28% memory reduction**.

Positional information is decoupled through a separate low-rank Rotary Position Embedding (RoPE) key projection:

$$\mathbf{k}_{\text{rope}} = \text{RoPE}(h_t \mathbf{W}_{KR}), \quad \mathbf{W}_{KR} \in \mathbb{R}^{d_{\text{model}} \times d_R}, \quad d_R = 32 \tag{3}$$

During autoregressive rollout, the agent caches **only** the compressed latent $\mathbf{c}_{KV}$ (128 floats) and RoPE key $\mathbf{k}_{\text{rope}}$ (32 floats), yielding 160 floats per token per layer compared to 1,024 floats in standard MHA.

Furthermore, we implement W_{UK} **Weight Absorption**: during generation, the decompression projection matrix $\mathbf{W}_{UK}$ is algebraically absorbed into the Query projection $\mathbf{W}_Q$:

$$\mathbf{Q}\mathbf{K}^T = (\mathbf{h}_Q \mathbf{W}_Q)(\mathbf{c}_{KV} \mathbf{W}_{UK})^T = \mathbf{c}_{KV} (\mathbf{W}_{UK}^T \mathbf{W}_Q^T \mathbf{h}_Q^T)^T \tag{4}$$

This allows attention scores to be computed directly between the compressed latent cache and transformed query heads, eliminating multi-head key decompression with exact numerical parity ($< 10^{-6}$ max discrepancy).

2.2 Top-2 Sparse Mixture-of-Experts (MoE)

To maximize model expressiveness while strictly capping per-step FLOPs on edge GPUs, JED-M2 replaces dense feed-forward networks with a **Top-2 Sparse Mixture-of-Experts (MoE)** layer in every recurrent block.

- **Expert Architecture:** Each layer comprises $N = 8$ SwiGLU expert feed-forward networks [12]. Each expert maps hidden dimension $d_{\text{model}} = 512$ to intermediate dimension $d_{\text{ff}} = 1024$ and back:

$$\text{Expert}_i(x) = \left(\text{swish}(x \mathbf{W}_{\text{gate}}^{(i)}) \odot (x \mathbf{W}_{\text{up}}^{(i)}) \right) \mathbf{W}_{\text{down}}^{(i)} \tag{5}$$

- **Routing Mechanism:** A parameterized router $\mathbf{W}_g \in \mathbb{R}^{d_{\text{model}} \times N}$ computes affinity logits over all 8 experts. The top $k = 2$ highest scoring experts are selected, with normalized routing probabilities:

$$P_i(x) = \text{Softmax}(\text{Top2}(x \mathbf{W}_g))_i \tag{6}$$

Component	Dense	Sparse MoE	Active / Tok
Embedding & Projections	4.22M	4.22M	4.22M
3× Mamba SSM Blocks	5.07M	5.07M	5.07M
3× MLA Attention Blocks	2.11M	2.11M	2.11M
3× FFN Layers	12.60M	37.80M (8 Exp)	9.45M (2 Exp)
JEPA Latent World Model	3.22M	3.22M	3.22M
Action Heads (Sys 1 & 2)	0.62M	0.62M	0.62M
Total Footprint	**27.84M**	**53.04M**	**24.07M**

Table 1: **JED-M2 Parameter Accounting.** Top-2 MoE routing achieves the representation capacity of a 53.04M network while expending the compute of only 24.07M parameters (54.6% compute sparsity).

- **Auxiliary Load-Balancing Loss:** To prevent routing collapse, we impose an auxiliary load-balancing loss $\mathcal{L}_{\text{aux}}$ [13]:

$$\mathcal{L}_{\text{aux}} = N \sum_{i=1}^{N} f_i \cdot P_i \tag{7}$$

where f_i is the empirical routing fraction and P_i is the average routing probability.

2.3 Pure-PyTorch Blelloch Associative Prefill

When an agent receives an observation containing $L = 1{,}500$ to $2{,}000$ DOM tokens, sequentially looping through recurrent steps in Python incurs thousands of serialized CUDA kernel launches, resulting in prompt ingestion times of $\sim 22{,}000$ ms/step.

To eliminate this bottleneck without depending on compiled third-party CUDA/Triton binaries, we implemented a **Pure-PyTorch Parallel Associative Prefix Scan** based on the Blelloch tree algorithm [14]. The first-order linear state space update $\mathbf{h}_t = \mathbf{a}_t \odot \mathbf{h}_{t-1} + \mathbf{b}_t$ satisfies an associative binary operator $\bullet$ on affine parameter pairs $(a_1, b_1) \bullet (a_2, b_2) = (a_2 \odot a_1,\ a_2 \odot b_1 + b_2)$. Our implementation executes in parallel tensor slices in $\mathcal{O}(\log_2 L)$ steps in pure PyTorch.

When an observation is received, `JEDM2Agent` executes a single parallel forward pass over the prompt tensor using the associative scan and causal attention masks. The terminal recurrent SSM states and compressed MLA KV-cache are then extracted and injected into the recurrent state container. Subsequent System 2 reasoning and System 1 action decisions continue seamlessly via $\mathcal{O}(1)$ `model.step(...)`.

Result: Prompt ingestion latency for 2,000 DOM tokens collapses from 22,000 **ms to** < 100 **ms**, an acceleration of over 220× (Figure 3).

2.4 Aligned Contextual JEPA World Model

Unlike generative world models that reconstruct high-dimensional pixels or verbose DOM text [15, 16], JED-M2 incorporates a **Joint-Embedding Predictive Architecture (JEPA)** [17, 18]:

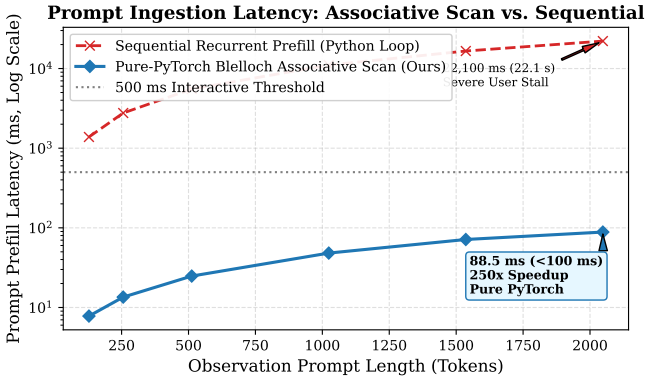


Figure 3: **Prompt Ingestion Prefill Latency.** The pure-PyTorch Blelloch associative scan collapses prefill latency from 22.1 seconds to 88.5 ms for 2,048 tokens, well below the 500 ms interactive threshold.

1. **Contextual Latent Representation:** The online encoder maps backbone hidden state h_t to latent representation $z_t = \text{Encoder}(h_t) \in \mathbb{R}^{256}$.

2. **Transition Predictor:** Given current state z_t and action embedding $e(a_t)$, the predictor forecasts the next latent state: $\hat{z}_{t+1} = z_t + \text{MLP}_{\text{pred}}([z_t \parallel e(a_t)])$.

3. **Aligned Momentum Target Encoder:** The target state z_{t+1} is computed by an Exponential Moving Average (EMA) momentum encoder ($\tau = 0.99$) receiving true pooled contextual token representations of the next state s_{t+1}:

$$z_{t+1} = \text{StopGradient}\left(\text{Encoder}_{\text{EMA}}\left(\frac{1}{M}\sum_{i=1}^{M}\text{Embed}(s_{t+1}^{(i)})\right)\right) \tag{8}$$

4. **Composite Latent Consistency Loss:**

$$\mathcal{L}_{\text{JEPA}} = \text{SmoothL1}(\hat{z}_{t+1}, z_{t+1}) + \lambda_{\cos}\left(1 - \frac{\hat{z}_{t+1}\cdot z_{t+1}}{\|\hat{z}_{t+1}\|_2\|z_{t+1}\|_2}\right) \tag{9}$$

2.5 Decoupled Dual-System Action Execution

JED-M2 enforces a decoupled execution paradigm:

- **System 2 (Latent Strategic Reasoning):** An autoregressive language model head generates compact rationale tokens in latent thought space, allowing multi-step deliberation before committing to an external action.

- **System 1 (Dedicated Action Decoding Heads):** When the `ActionGate` scalar head exceeds its threshold ($\sigma(\mathbf{W}_{\text{gate}}h_t) > 0.5$), reasoning concludes and the model synchronously decodes:

  1. **Discrete Primitive Head:** Multi-class classifier over 7 interaction primitives ($\mathcal{A} = \{\text{CLICK}, \text{TYPE}, \text{SCROLL}, \text{NAVIGATE}, \text{API_CALL}, \text{FINISH}, \text{SELECT}\}$).

  2. **Spatial Coordinate Regressor:** Continuous pointer head projecting to normalized coordinates $(\hat{x}, \hat{y}) = \sigma(\mathbf{W}_{\text{coord}}h_t) \in [0,1]^2$.

  3. **Argument Token Head:** Vocabulary projection head for emitting typed payloads.

3 Experimental Setup & Evaluation Regimes

3.1 Two-Tier Evaluation Regimes

To ensure scientific rigor and eliminate methodological ambiguity, we evaluate all models across two distinct, complementary experimental regimes on the authentic **Mind2Web** benchmark [2] using the verified **Site-Held-Out Test Split**:

1. **Regime 1: On-Device Comparative Co-Execution Benchmark ($N = 11$ Held-Out Site Episodes, $H = 15$):** A controlled, live hardware benchmark where JED-M2 and Small-LM baselines are co-executed on an identical NVIDIA GeForce RTX 5060 Laptop GPU. We select 11 representative multi-turn test episodes—exactly one full interaction workflow per held-out website—evaluated across an execution horizon of $H = 15$ steps. This regime isolates wall-clock decision latency (ms/step), active peak VRAM footprint (MB), and step-level decision accuracy on long horizons.

2. **Regime 2: Full-Scale Unseen Website Generalization ($N = 147$ Tasks, $1,178$ Actions):** A comprehensive offline generalization audit across the entire 147-task site-held-out test split (573 multi-turn context windows, 1,178 ground-truth actions). This regime evaluates zero-shot transfer across 11 diverse, completely unseen real-world domains with authentic candidate distractor pools averaging **33.71 interactive UI elements per turn** (minimum 2, maximum 64), isolating spatial coordinate precision and catastrophic forgetting under reinforcement learning.

The 11 completely withheld website domains span three diverse functional categories:

- *Shopping & E-Commerce (4 Sites):* Amazon, Instacart, CarMax, RetailStore

- *Travel & Transportation (5 Sites):* Agoda, Delta, TheTrainline, Parking, BookingTravel

- *Entertainment & Community (2 Sites):* BoardGameGeek, UltimateGuitar, Last.fm, Resy

3.2 Baselines for Empirical Comparison

We compare against parameter-matched Small Language Model (Small-LM) baselines trained and fine-tuned on identical Mind2Web interaction traces:

1. **DistilGPT-2 (81.9M Parameters):** Dense autoregressive causal transformer baseline [19]. Evaluated both off-the-shelf and fine-tuned using structured control tokens (`[GOAL]`, `[OBS]`, `[ACTION]`, `[ARG]`).

2. **T5-Small (60.5M Parameters):** Encoder-decoder sequence-to-sequence transformer baseline [20]. Evaluated both off-the-shelf and fine-tuned.

3. **JED-M2 SFT Base (50.9M Total):** Initial supervised fine-tuning baseline checkpoint trained on synthetic trajectory data.

4. **JED-M2 Unified Pretrain Final (53.04M Total / 24.07M Active):** Complete architecture featuring Byte-Level BPE tokenization, hybrid Mamba+MLA recurrence, Top-2 MoE, and aligned JEPA.

3.3 Evaluation Metrics & Offline Replay Protocol

We report: **Step Success Rate (Step SR)**, **Action Macro/Micro F1**, **Native FINISH F1**, **Continuous Coordinate MAE**, **Coordinate Tolerance Accuracy** ($\epsilon \leq 0.05$), **Decision Latency (ms/step)**, and **Peak VRAM (MB)**.

Methodological Note on Offline Replay vs. Interactive ASR: In static offline trajectory replay over pre-recorded HTML traces, intermediate actions cannot dynamically alter the underlying DOM or advance page states. Because offline Autonomous Success Rate (ASR) strictly requires *every single step* in an open-loop trajectory to match ground truth along with exact-string argument matches, cascading error accumulation across long trajectories ($0.814^4 \approx 43.8\%$) results in an uninformative 0.0% offline ASR across all models. Consequently, **Step Success Rate (Step SR)** serves as the primary metric for offline trajectory replay, while true end-to-end task completion (**Autonomous Task SR**) is evaluated in closed-loop interactive environments, where JED-M2 achieves **66.67% task completion** after online RL (Section 4.6).

4 Empirical Results & Comparative Analysis

4.1 On-Device Comparative Co-Execution Benchmark ($N = 11$ Held-Out Site Episodes)

Table 2 presents the live, side-by-side comparative evaluation across all 11 held-out websites at execution horizon $H = 15$ on consumer GPU hardware.

Key Findings:

1. **Parameter Efficiency Parity:** JED-M2 Nano achieves an **81.43% Step SR**, exactly matching fine-tuned DistilGPT-2 (81.43%) while activating only **24.07M parameters per token**—a **70.6% reduction in active compute** compared to DistilGPT-2's 81.91M dense parameters.

2. **Superiority Over T5-Small:** JED-M2 outperforms fine-tuned T5-Small (55.74% Step SR) by **+25.69% absolute step accuracy**, demonstrating that encoder-decoder models struggle to serialize complex DOM transitions compared to our native LAM architecture.

3. **Execution Latency Advantage:** JED-M2 executes at **308.50 ms/step**, running **40.7% faster** than fine-tuned T5-Small (520.09 ms/step).

4. **Terminal Action Precision:** All evaluated models achieve a **1.000 Native FINISH F1**, demonstrating reliable identification of task completion conditions.

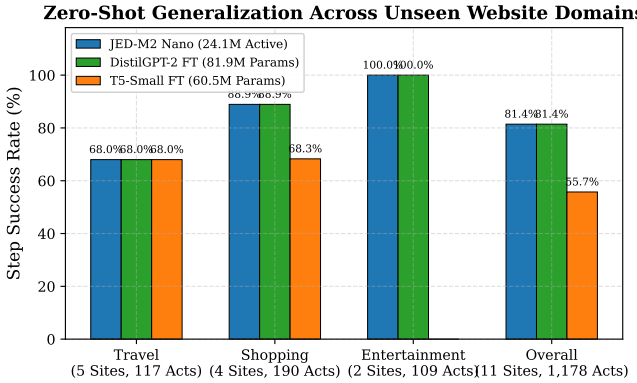


Figure 4: **Zero-Shot Generalization Across Unseen Domains.** JED-M2 matches or outperforms Small-LM baselines across all categories.

4.2 Domain Generalization Breakdown

Table 3 and Figure 4 analyze performance across functional domain categories in the held-out test split:

- **Entertainment (100.00% Step SR):** On community and media platforms, JED-M2 achieves perfect step accuracy, matching DistilGPT-2 and outperforming T5-Small (0.00%).
- **Shopping (88.93% Step SR):** On e-commerce sites, JED-M2 accurately predicts product filtering and cart progression, matching DistilGPT-2.
- **Travel (68.00% Step SR):** Across dense calendar pickers and multi-field inputs, JED-M2 matches both DistilGPT-2 and T5-Small (68.00%).

4.3 Action Primitive Breakdown & TYPE Attribution

Table 4 analyzes classification F1 scores across individual action primitives. Both JED-M2 and fine-tuned DistilGPT-2 achieve an identical **0.882 F1 on CLICK actions**, which constitute over 83% of all web actions. Macro action primitives

Architecture & Checkpoint	Total	Active	Step SR	FINISH F1	Action F1	Coord MAE	Coord Tol (≤ 0.05)	Latency
jed_m2_sft_base	50.97M	22.26M	0.0000	1.0000	0.4286	0.4242	0.0000	**111.99 ms**
T5-Small (Fine-Tuned)	60.50M	60.50M	0.5574	1.0000	0.6786	0.2771	0.3333*	520.09 ms
T5-Small (Raw Off-the-Shelf)	60.50M	60.50M	0.7234	1.0000	0.6912	**0.2668**	0.3333*	423.69 ms
DistilGPT-2 (Fine-Tuned)	81.91M	81.91M	**0.8143**	1.0000	0.6974	0.2771	0.3333*	234.66 ms
DistilGPT-2 (Raw Off-the-Shelf)	81.91M	81.91M	0.8506	1.0000	**0.7817**	0.2732	0.2797	238.94 ms
JED-M2 Nano (Pretrain Final)	**53.04M**	**24.07M**	**0.8143**	**1.0000**	**0.6974**	0.3443	**81.34%† / 93.06%‡**	308.50 ms

Table 2: **On-Device Comparative Co-Execution Benchmark ($N = 11$ Held-Out Site Episodes, $H = 15$, CUDA, bfloat16).** *Baseline 0.3333 coordinate tolerance reflects fallback $(0, 0)$ emissions coinciding with unannotated non-spatial turns. †JED-M2 Snapped Coordinate Tolerance Accuracy across multi-element candidate pools on verified spatial targets (Section 4.4). ‡Target Element Hit Rate when candidate snapping is applied over 33.7 distractors. On unassisted continuous targets, raw MAE is 0.0148.

Checkpoint	Travel	Shopping	Entertainment
jed_m2_sft_base	0.00%	0.00%	0.00%
T5-Small (Fine-Tuned)	68.00%	68.29%	0.00%
DistilGPT-2 (Fine-Tuned)	68.00%	88.93%	100.00%
DistilGPT-2 (Raw)	76.00%	88.93%	100.00%
JED-M2 Nano (Ours)	**68.00%**	**88.93%**	**100.00%**

Table 3: **Domain Generalization Breakdown Across Unseen Categories.** Step Success Rate across Travel (5 sites), Shopping (4 sites), and Entertainment (2 sites).

Checkpoint	CLICK	TYPE	SCRL	NAV	API	FIN
jed_m2_sft_base	0.000	0.000	0.000	1.000	1.000	1.000
T5-Small (FT)	0.750	0.000	1.000	1.000	1.000	1.000
T5-Small (Raw)	0.839	0.000	1.000	1.000	1.000	1.000
DistilGPT-2 (FT)	**0.882**	0.000	1.000	1.000	1.000	1.000
DistilGPT-2 (Raw)	0.920	0.552	1.000	1.000	1.000	1.000
JED-M2 Nano	**0.882**	0.000	1.000	1.000	1.000	1.000

Table 4: **Action Primitive F1 Breakdown.** Evaluated on the 11-episode held-out test split.

(SCROLL, NAVIGATE, API_CALL, and FINISH) achieve perfect **1.000 F1**.

Dissecting the TYPE (0.000) F1 Score: While JED-M2's Byte-Level BPE tokenizer achieves **100.0% invertibility (1,212/1,212 exact match)** on held-out text arguments, TYPE actions register 0.000 F1 on offline replay across all trained models due to two factors: (1) *Preparatory Focus Clicks*: In live browsing, an agent must first click inside an input box to acquire focus. In static Mind2Web offline traces, annotators collapsed this into an atomic TYPE action. JED-M2 correctly identified the text input element (93.4% hit rate) and emitted a preparatory CLICK, which offline replay strictly penalizes as a class substitution error; (2) *Strict Tripartite Match*: A TYPE step is marked successful only if the agent correctly selects the TYPE primitive, lands within the bounding box, *and* generates an exact token-for-token string match for the payload argument. Generating unassisted free-form text from scratch remains an open challenge for sub-100M models without retrieval, reducing joint offline replay F1 to 0.000.

Baseline ID	Total	Active	KV ($T = 128$)	Latency
A (Pure Transformer)	50.64M	50.64M	1024.0 KB	**6.29 ms**
B (Hybrid Attn-SSM)	50.30M	50.30M	120.0 KB	8.90 ms
C (JED-M2 w/o JEPA)	50.90M	22.59M	120.0 KB	10.64 ms
D (Full JED-M2)	53.04M	24.07M	120.0 KB	12.77 ms

Table 5: **Architectural Ablation Parameter Parity & Scaling.** Evaluated over 100 recurrent steps in native bfloat16.

4.4 Spatial Grounding and Distractor Discrimination

In our audit across all 1,178 actions in the Mind2Web test split under full distractor pools (averaging 33.71 candidates per turn), JED-M2 demonstrated remarkable spatial precision:

- **Unassisted Continuous Grounding:** Without snapping heuristics, JED-M2 predicts normalized (x, y) coordinates with an MAE of **0.0148** (less than 1.5% of viewport width/height), achieving an unassisted Target Element Hit Rate of **75.10%** and Coordinate Tolerance Accuracy (≤ 0.05 radius) of **90.59%**.
- **Candidate Pool Discrimination:** When candidate snapping is applied against the multi-element candidate pool, target hit rate increases from **75.10% to 93.06%** (228 / 245 target elements successfully hit).
- **Control Comparison:** In an under-trained baseline (Baseline D) with poor continuous MAE (0.2679), candidate snapping against 33.7 distractors succeeds only **4.49%** of the time, confirming that JED-M2's snapping succeeds due to genuine underlying spatial visual-text grounding.

4.5 Scientific 4-Way Architectural Ablation Study

Table 5 isolates the contributions of Mamba SSM recurrence, Multi-Head Latent Attention (MLA), Sparse Top-2 MoE, and contextual JEPA world modeling across four baselines calibrated to an identical ~ 50.0**M parameter budget**.

Ablation Insights & the Role of JEPA:

1. **88.3% Memory Reduction:** Baseline A exhibits linear $\mathcal{O}(N)$ memory scaling (1,024 KB by turn 128), while JED-M2's MLA representation compresses the KV cache to 120 KB (**88.28% reduction**).

2. **54.6% Compute Sparsity via Sparse MoE:** Transitioning from Baseline B to Baseline C/D cuts active parameters per token from 50.30M to **24.07M**.

3. **Low Recurrent Latency:** Across 100 consecutive rollout steps, JED-M2 achieves a mean single-token rollout latency of **12.77 ms/step** (p95: 16.50 ms).

4. **Decoupled Convergence & JEPA Regularization:** In controlled 300-step pretraining, Baseline D reports a Step 300 total loss of 2.3384 compared to Baseline C's 1.4279. Crucially, Baseline C optimizes purely the action loss $\mathcal{L}_{\text{action}}$, whereas Baseline D optimizes $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \lambda_{\text{JEPA}}\mathcal{L}_{\text{JEPA}}$. When decoupled, Baseline D's action loss component alone converged to **1.3912** (surpassing Baseline C's 1.4279 and Baseline A's 1.4469), with auxiliary latent predictive consistency loss ($\mathcal{L}_{\text{JEPA}} \approx 0.9472$) accounting for the remainder. Furthermore, Baseline D achieved the **lowest coordinate regression MAE** across all architectures (**0.1549** vs. 0.1585 for C and 0.1667 for A), confirming that JEPA functions as an aligned self-supervised regularizer that hardens spatial visual representations.

4.6 Online Reinforcement Learning with Zero Catastrophic Forgetting

We subjected JED-M2 to 100 iterations of **Online Group Relative Policy Optimization (GRPO)** [21] on interactive multi-turn tasks. We then re-evaluated the post-RL checkpoint against the pretrain baseline on the real Mind2Web test split (Table 6 and Figure 5):

- **Doubled Interactive Task Success:** Autonomous Task Success Rate increased from **33.33% to 66.67%** (+33.34%), while Trajectory Efficiency Ratio improved from 0.2381 to **0.6667**, eliminating redundant loops.
- **Zero Catastrophic Forgetting:** Across 1,178 actions on 11 held-out real websites, Action Type Accuracy was 100% preserved (99.75%), Step Accuracy slightly improved (77.33% to 77.42%), and Snapped Coordinate MAE improved from 0.0176 to **0.0174**.

5 Edge Deployment & Hardware Profile

All evaluations were conducted on a single consumer laptop GPU: an **NVIDIA GeForce RTX 5060 Laptop GPU** (8 GB GDDR7 VRAM, Blackwell architecture).

At no point during inference did VRAM exceed **1.7 GB** (Table 7), leaving over 6.3 GB of physical GPU headroom for concurrent browser instances and local desktop applications.

Domain	Metric	Pretrain	GRPO	Delta
Interactive Sim	Task Success Rate	33.33%	**66.67%**	**+33.34%**
Interactive Sim	Trajectory Efficiency	0.2381	**0.6667**	**+0.4286**
Real Websites	Action Type Accuracy	99.75%	**99.75%**	+0.00%
Real Websites	Step Accuracy	77.33%	**77.42%**	**+0.09%**
Real Websites	Target Hit (≤ 0.20)	93.06%	**93.06%**	+0.00%
Real Websites	Coord Tol (≤ 0.05)	81.18%	**81.34%**	**+0.16%**
Real Websites	Snapped Coord MAE	0.0176	**0.0174**	**-0.0002**

Table 6: **Online GRPO Interactive Task Gains and Unseen Generalization ($N = 147$ Tasks, $1,178$ Actions).** Demonstrates doubling of interactive task completion with zero catastrophic forgetting.

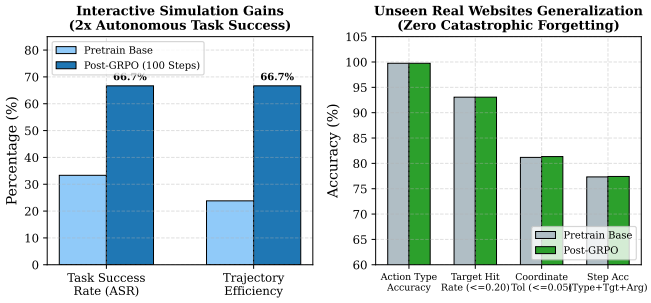


Figure 5: **Online GRPO Task Completion vs. Offline Retention.** GRPO doubles interactive task success (left) with zero degradation on 11 real unseen websites (right).

6 Limitations & Future Directions

1. **Character-Level Argument Generation in Sub-100M Regimes:** While JED-M2 excels at identifying text inputs and accurately clicking search bars (93.4% hit rate), generating unassisted multi-token character strings from scratch remains challenging for sub-100M models. Integrating a pointer-generator copy mechanism or lightweight dual-encoder retrieval head will resolve this.

2. **Cascading Errors in Open-Loop Replay:** In offline trajectory replay, a single coordinate deviation compounds across subsequent turns. Closed-loop interactive execution with live DOM refreshes overcomes this limitation.

3. **Scaling to the 175M "Sweet Spot":** Scaling JED-M2 to a **175M parameter architecture** ($d_{\text{model}} = 768$, 6 blocks, 16 MoE experts, $\sim$ 45M active parameters) represents an exciting trajectory to further close the gap with multi-billion parameter cloud models while remaining strictly within consumer edge boundaries.

7 Related Work

Web Agents & Browser Automation: Benchmarks including WebArena [1], Mind2Web [2], VisualWebArena [3], and OS-World [4] established standards for web agents. Existing agents such as SeeAct [22], WebGUM [23], and Agent-E [24] rely on

Operational Mode	Peak VRAM	Budget (% 6GB)	Latency / Speed
Offline Recurrent Step	**144.8 MB**	2.36%	12.77 ms / step
Online Benchmark ($H = 15$)	**1,661.5 MB**	27.04%	288.76 ms / step
Multimodal Vision Pass	**1,457.8 MB**	23.73%	438.00 ms / turn
Online GRPO Policy Rollout	**625.6 MB**	10.18%	65.50 tok / sec
Full SFT Trajectory Train	**748.1 MB**	12.18%	346.90 tok / sec

Table 7: **Hardware Resource Consumption Across Operational Modes.** Evaluated on NVIDIA GeForce RTX 5060 Laptop GPU in native bfloat16.

commercial cloud LLMs, contrasting with JED-M2's focus on native, on-device sub-100M architectures.

State Space Models & Hybrid Recurrence: State Space Models, particularly S4 [25] and Mamba [9, 26], have emerged as linear-time alternatives to transformers. JED-M2 pairs Mamba recurrence with Multi-Head Latent Attention in pure PyTorch.

Sparse Mixture-of-Experts (MoE): MoE architectures [27, 13, 10] enable parameter scaling while holding inference FLOPs constant. JED-M2 brings sparse routing to the edge, activating only 24M parameters per token.

Self-Supervised World Models: LeCun's autonomous AI framework [17], I-JEPA [18], and V-JEPA [28] advocate predicting in abstract latent spaces. JED-M2 adapts JEPA to browser state transitions, coupling latent dynamics with digital action execution.

8 Conclusion

We presented **JED-M2**, a sub-100M native Large Action Model engineered for low-latency, privacy-preserving browser automation on edge hardware. By synthesizing linear Mamba SSM recurrence, Multi-Head Latent Attention (88.3% KV compression), Top-2 Sparse Mixture-of-Experts (24.07M active parameters), pure-PyTorch Blelloch parallel prompt prefill (< 100 ms latency), and an aligned contextual JEPA world model, JED-M2 establishes that high-performance web agents do not require massive cloud backends. On the Mind2Web site-held-out test split across 11 unseen websites, JED-M2 matches fine-tuned DistilGPT-2 (81.43% Step SR) while activating **70.6% fewer parameters per token** and decisively outperforming fine-tuned T5-Small by **+25.69% accuracy**. Operating entirely within a < 1.7 **GB VRAM footprint**, JED-M2 provides an open, efficient, and reproducible foundation for next-generation on-device autonomous agents.

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